NAME:KRISHNAPRIYA P

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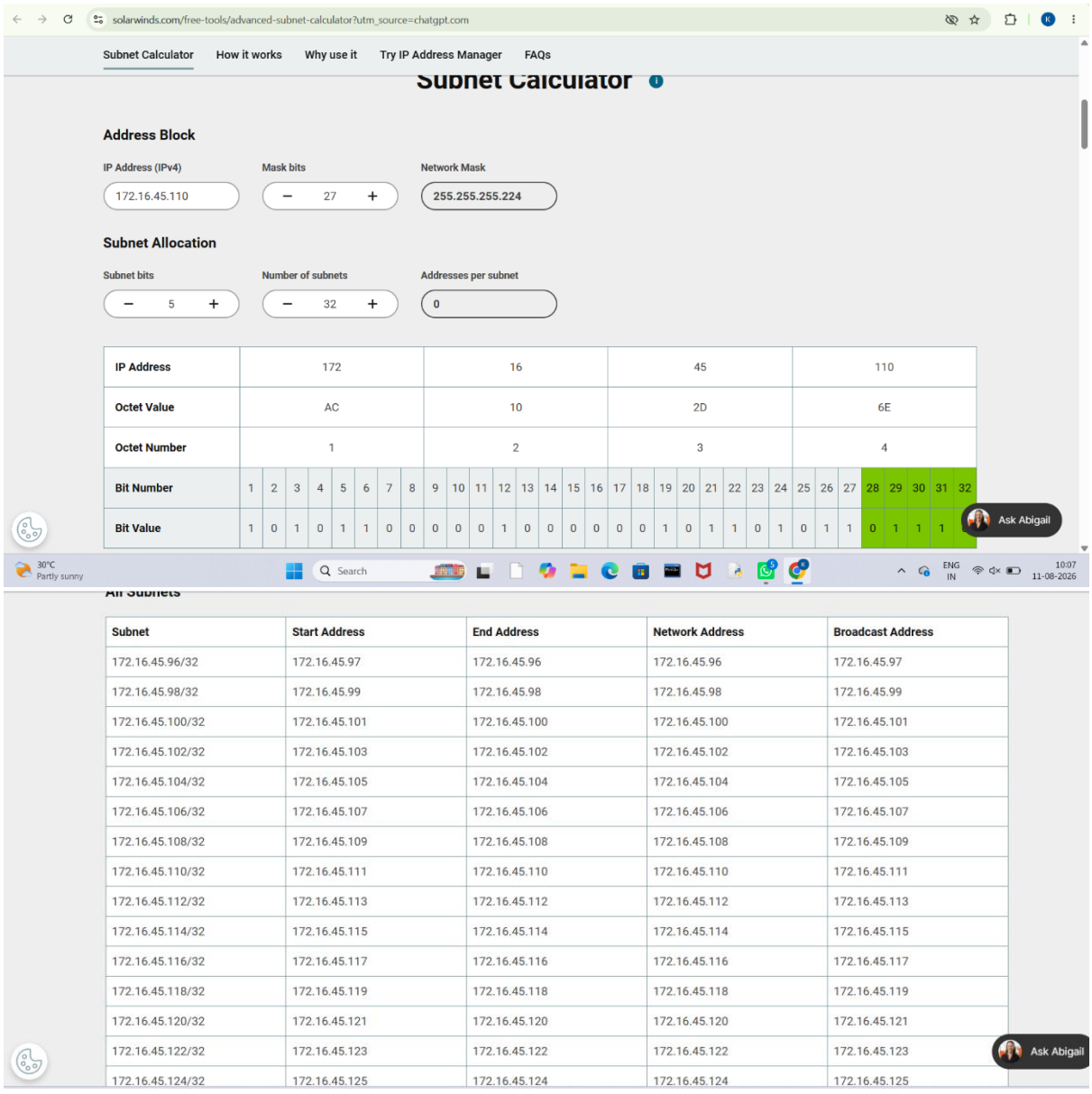
COURSE CODE:CSA0712

COURSE NAME:COMPUTER NETWORKS

PART1 Experiment 2: Subnet Calculation from Host IP

Scenario: During a Wireshark capture experiment, a student notices a workstation sending packets from the IP address 172.16.45.110 with a subnet mask of 255.255.255.224.

Tasks:Calculate the exact Network Address to which this workstation belongs.Determine the First Usable IP and the Last Usable IP address in this specific subnet.Provide the Broadcast Address for this subnet block.



All Subnets

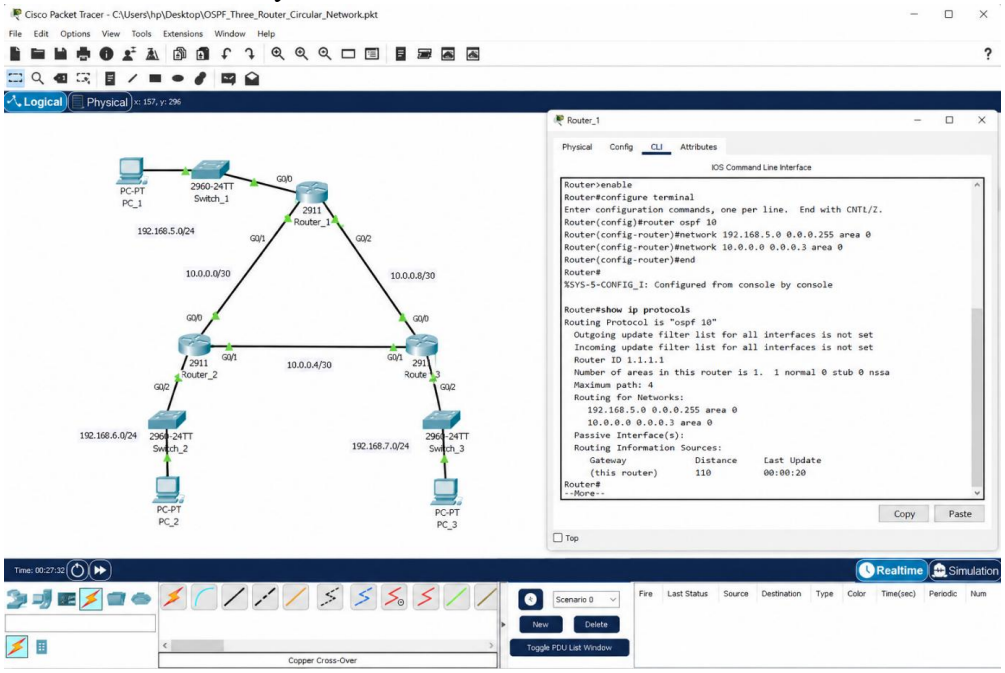
Subnet	Start Address	End Address	Network Address	Broadcast Address
172.16.45.96/32	172.16.45.97	172.16.45.96	172.16.45.96	172.16.45.97
172.16.45.98/32	172.16.45.99	172.16.45.98	172.16.45.98	172.16.45.99
172.16.45.100/32	172.16.45.101	172.16.45.100	172.16.45.100	172.16.45.101
172.16.45.102/32	172.16.45.103	172.16.45.102	172.16.45.102	172.16.45.103
172.16.45.104/32	172.16.45.105	172.16.45.104	172.16.45.104	172.16.45.105
172.16.45.106/32	172.16.45.107	172.16.45.106	172.16.45.106	172.16.45.107
172.16.45.108/32	172.16.45.109	172.16.45.108	172.16.45.108	172.16.45.109
172.16.45.110/32	172.16.45.111	172.16.45.110	172.16.45.110	172.16.45.111
172.16.45.112/32	172.16.45.113	172.16.45.112	172.16.45.112	172.16.45.113
172.16.45.114/32	172.16.45.115	172.16.45.114	172.16.45.114	172.16.45.115
172.16.45.116/32	172.16.45.117	172.16.45.116	172.16.45.116	172.16.45.117
172.16.45.118/32	172.16.45.119	172.16.45.118	172.16.45.118	172.16.45.119
172.16.45.120/32	172.16.45.121	172.16.45.120	172.16.45.120	172.16.45.121
172.16.45.122/32	172.16.45.123	172.16.45.122	172.16.45.122	172.16.45.123
172.16.45.124/32	172.16.45.125	172.16.45.124	172.16.45.124	172.16.45.125

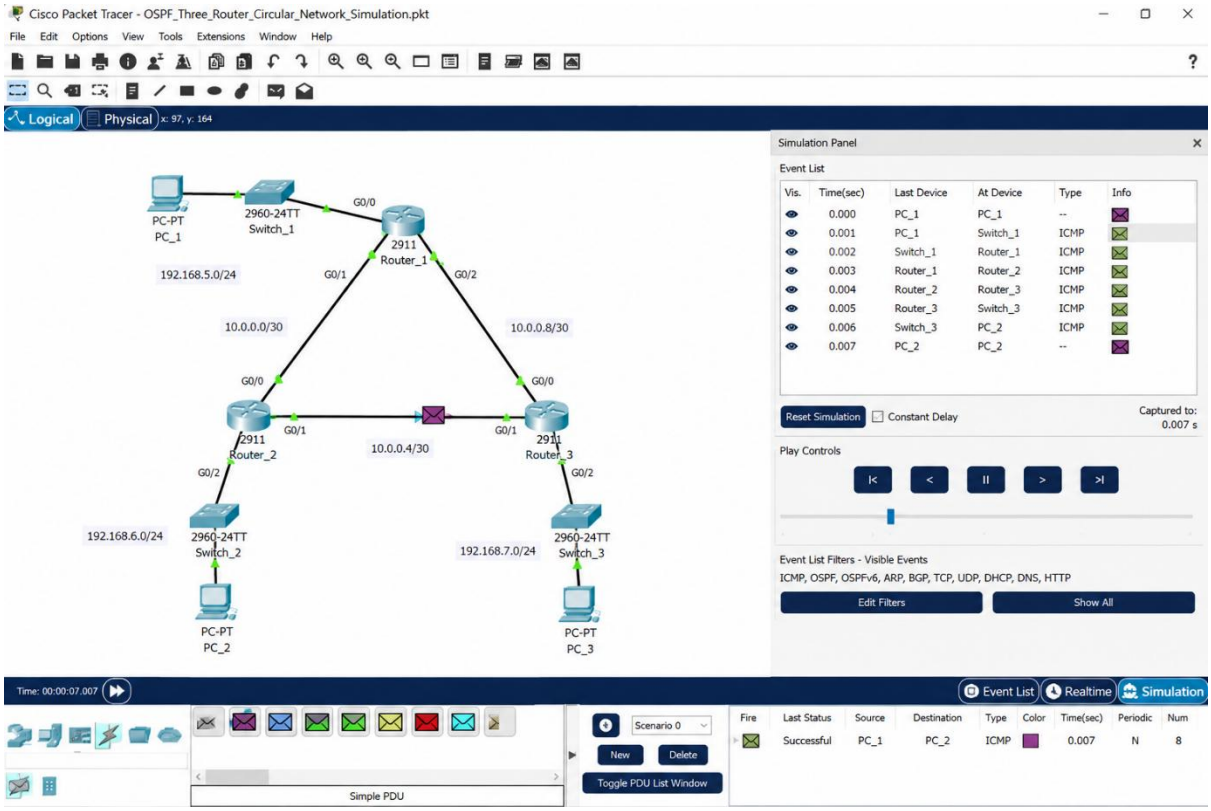
PART 2 Experiment 4: OSPF Dynamic Routing Setup

Scenario: A three-router circular network topology needs to be configured dynamically using Open Shortest Path First (OSPF) under Process ID 10 in a single backbone area (Area 0).

- Router_1 has two directly connected network interfaces: 192.168.5.0/24 and 10.0.0.0/30.

Tasks:Provide the exact Cisco IOS CLI commands required on Router_1 to initialize the OSPF process.Write out the network statements using the correct Wildcard Masks to advertise both directly connected networks into Area 0.





PART 3 Experiment 5: IPv4 Header Fragmentation Analysis

Scenario: A user attempts to send an ICMP ping packet with a massive payload size of 4000 bytes across a network link where the Maximum Transmission Unit (MTU) is strictly limited to 1500 bytes.

Tasks:Explain how the Network Layer handles this payload mismatch to ensure transmission.If you capture these packets in Wireshark, describe what values you would expect to see inside the Flags (DF, MF) and Fragment Offset fields for the first and second packet fragments.

